

Nutrition & Preventive Medicine

Evidence-Based Dietary Guidelines, Screening & Health Promotion

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1. Macronutrients

Macronutrients — carbohydrates, proteins, and fats — are the primary sources of energy and structural components for the human body. Dietary guidelines recommend specific proportions to support optimal health and disease prevention.

Macronutrient	Energy Density	Recommended Intake (% total energy)	Primary Sources
Carbohydrates	4 kcal/g	45–65%	Wholegrains, legumes, fruits, vegetables
Proteins	4 kcal/g	10–35%	Lean meats, fish, eggs, dairy, legumes, nuts
Total Fats	9 kcal/g	20–35%	Olive oil, nuts, avocado, fatty fish
Saturated Fats	9 kcal/g	< 10% (< 7% if high CV risk)	Reduce: red meat, butter, cream, coconut oil
Trans Fats	9 kcal/g	< 1% (minimise entirely)	Eliminate: partially hydrogenated oils, pastries

1.1 Dietary Fibre

Dietary fibre is a non-digestible carbohydrate fraction that exerts multiple health benefits: improved glycaemic control, reduced LDL cholesterol, bowel regularity, satiety, and colorectal cancer risk reduction.

- Recommended intake: 25–38 g/day (adults); most populations consume only 15–18 g/day
- Soluble fibre (oats, psyllium, legumes): Reduces LDL-C, blunts post-prandial glucose
- Insoluble fibre (wheat bran, vegetables): Accelerates intestinal transit, prevents constipation
- High-fibre diets associated with reduced risk of T2DM, coronary heart disease, colorectal cancer

2. Micronutrients — Key Deficiencies

Nutrient	Deficiency Syndrome	At-Risk Groups	Dietary Sources
Iron	Iron-deficiency anaemia (fatigue, pallor, koilonychia)	Women of childbearing age, vegetarians, infants	Red meat, liver, legumes, fortified cereals, spinach
Vitamin B12	Megaloblastic anaemia, subacute combined degeneration of cord	Strict vegans/vegetarians, pernicious anaemia, elderly, metformin users	Meat, fish, eggs, dairy, fortified plant milks
Vitamin D	Rickets (children), osteomalacia (adults), immune dysfunction	Dark-skinned individuals, elderly, veiled women, housebound, northern latitudes	Oily fish, fortified dairy, fortified cereals; UV sunlight (primary source)
Folate (B9)	Neural tube defects (foetal), megaloblastic anaemia	Women of childbearing age, pregnancy, alcohol excess	Leafy vegetables, legumes, fortified bread; supplement 400 µg preconception–12 wks
Iodine	Hypothyroidism, goitre, cretinism (congenital)	Inland communities, vegans, pregnancy	Seafood, dairy, iodised salt, seaweed
Calcium	Osteoporosis, tetany, rickets	Postmenopausal women, vegans, lactose intolerant	Dairy, fortified plant milks, tofu, canned fish with bones

3. Cancer Screening Programmes

Screening aims to detect cancer or pre-cancerous lesions in asymptomatic individuals at a stage when treatment is most effective. Screening programmes must balance sensitivity, specificity, cost, and potential harms (overdiagnosis, false positives, procedure risk).

Cancer	Screening Test	Target Population	Frequency
Breast	Mammography	Women 40–74 (50–74 high priority)	Every 1–2 years
Cervical	Pap smear / HPV DNA test	Women 21–65 (3 yrs after onset of sexual activity)	Every 3 yrs (Pap) or 5 yrs (co-test)
Colorectal	Faecal immunochemical test (FIT)	Age 45–75 years (average risk)	Annual FIT; colonoscopy q10yr

Cancer	Screening Test	Target Population	Frequency
Lung	Low-dose CT (LDCT)	Age 50–80, ≥20 pack-year, current or quit < 15 yrs	Annual
Prostate	PSA + shared decision-making	Men 55–69 (after discussion)	Every 1–2 years if elected

4. Vaccination Schedule (Adult)

Immunisation is one of the most cost-effective preventive health interventions. The following summarises key adult vaccines recommended in most national programmes.

Vaccine	Indication / Target	Schedule
Influenza	All adults ≥6 months annually	Single dose; annual; autumn before flu season
COVID-19	All adults; high-risk groups prioritised	Primary series + boosters per national guidance
Tetanus-Diphtheria (Td)	All adults	Booster every 10 years; Tdap once (contains pertussis)
MMR (Measles-Mumps -Rubella)	Seronegative adults; HCW	2 doses 4 weeks apart if not immune
Hepatitis B	All non-immune adults; HCW; risk groups	3-dose series at 0, 1, 6 months
Pneumococcal (PCV20/PPSV23)	Age ≥65; immunocompromised; chronic illness	Per age/risk-based schedule
HPV (Gardasil 9)	Age 9–26 yrs; shared decision 27–45 yrs	2 doses (< 15 yrs) or 3 doses (≥15 yrs)
Shingles (Zoster — Shingrix)	Adults ≥50 years	2 doses 2–6 months apart; preferred over live vaccine

5. Lifestyle Medicine — Key Recommendations

5.1 Physical Activity

- Adults: ≥150 min/week moderate-intensity aerobic activity OR ≥75 min/week vigorous-intensity
- Muscle-strengthening activities on ≥2 days/week (major muscle groups)
- Reduce prolonged sedentary time — break sitting every 30 minutes

- Benefits: Reduced CV mortality, T2DM, depression, cancer, dementia risk; improved bone density

5.2 Alcohol & Tobacco

- Alcohol: No safe level; if consumed, < 14 units/week spread over ≥3 days (UK); > 2 standard drinks/day = harmful
- Tobacco: No safe level; smoking cessation is the single most effective preventive intervention
- Smoking cessation support: Brief advice + pharmacotherapy (varenicline, NRT, bupropion) increases quit rates 3–5x
- E-cigarettes: Less harmful than combustible tobacco but not risk-free; not recommended for non-smokers

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